Analysis

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TPOT with constrained search space versus ensemble

2000 total evals

Version 1

Stats Testing

```
## # A tibble: 26 x 9
##    task_id .y.   group1            group2    n1    n2 statistic       p p.signif
##      <int> <chr> <chr>             <chr>  <int> <int>     <dbl>   <dbl> <chr>
##  1    2073 auroc ensemble_random_~ indiv~    21    21      215  1.61e-4 ***
##  2  146818 auroc ensemble_random_~ indiv~    21    21      216. 5.43e-4 ***
##  3  146820 auroc ensemble_random_~ indiv~    21    21      192  6.28e-3 **
##  4  168350 auroc ensemble_random_~ indiv~    21    21      228  4.77e-6 ****
##  5  168757 auroc ensemble_random_~ indiv~    21    21      199  2.48e-3 **
##  6  168784 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
##  7  168911 auroc ensemble_random_~ indiv~    21    21      214  1.97e-4 ***
##  8  190137 auroc ensemble_random_~ indiv~    21    21      228  4.77e-6 ****
##  9  190146 auroc ensemble_random_~ indiv~    21    21      146  3.04e-1 ns
## 10  190411 auroc ensemble_random_~ indiv~    21    21      213  2.41e-4 ***
## 11  359955 auroc ensemble_random_~ indiv~    21    21      196  5.42e-3 **
## 12  359956 auroc ensemble_random_~ indiv~    21    21      179  2.63e-2 *
## 13  359958 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 14  359959 auroc ensemble_random_~ indiv~    21    21      216  1.31e-4 ***
## 15  359960 auroc ensemble_random_~ indiv~    21    21       23  5.29e-1 ns
## 16  359962 auroc ensemble_random_~ indiv~    21    21      190  8.01e-3 **
## 17  359963 auroc ensemble_random_~ indiv~    21    21      220  5.25e-5 ****
## 18  359964 auroc ensemble_random_~ indiv~    21    21      163  1.03e-1 ns
## 19  359965 auroc ensemble_random_~ indiv~    21    21      166. 2.28e-2 *
## 20  359968 auroc ensemble_random_~ indiv~    21    21      192  6.28e-3 **
## 21  359969 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 22  359970 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 23  359971 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 24  359972 auroc ensemble_random_~ indiv~    21    21      223  2.38e-5 ****
## 25  359974 auroc ensemble_random_~ indiv~    21    21      230  1.91e-6 ****
## 26  359975 auroc ensemble_random_~ indiv~    21    21      162. 1.06e-1 ns
```

Medians

```
## # A tibble: 26 x 3
##    task_id individual_tpot ensemble_random_100
##      <int>           <dbl>               <dbl>
##  1    2073           0.878               0.886
##  2  146818           0.931               0.938
##  3  146820           0.995               0.996
##  4  168350           0.961               0.965
##  5  168757           0.795               0.800
##  6  168784           0.965               0.968
##  7  168911           0.869               0.871
##  8  190137           0.916               0.926
##  9  190146           0.966               0.966
## 10  190411           0.918               0.921
## 11  359955           0.731               0.746
## 12  359956           0.934               0.934
## 13  359958           0.935               0.947
## 14  359959           0.751               0.755
## 15  359960           1                   1
## 16  359962           0.843               0.853
## 17  359963           0.995               0.996
## 18  359964           0.993               0.994
## 19  359965           1.00                1.00
```

```
## 20  359968          0.919             0.923
## 21  359969          0.845             0.851
## 22  359970          0.906             0.925
## 23  359971          0.997             0.997
## 24  359972          0.990             0.993
## 25  359974          0.818             0.868
## 26  359975          0.994             0.995
```